notesthemejambro.sty

Feature Demo

v1.0 — *March 23, 2026*

Contents

I Typography and Fonts

I.A Body Text

The default font is **Cabin** (sans-serif), loaded via `[sfdefault]{cabin}`. All body text, headings, and captions use this family. Math is rendered in the standard Computer Modern math fonts alongside Cabin.

A short paragraph: The quick brown fox jumps over the lazy dog. Numbers: 0123456789. Greek: $\alpha, \beta, \gamma, \delta, \epsilon$. Operators: $\sum_{i=1}^{n} x_i$, $\int_0^\infty e^{-x}\,dx$.

I.B Handwriting Font

The Augie font is available for hand-drawn effects:

- This sentence is set in the Augie handwriting font.
- **This sentence uses handBold — Bold Augie via PDF stroke literal.**
- Mixed: regular text alongside handwritten words in the same line.

I.C Pencil Underlines

Default underline uses the `carandache` color (goldenrod). A custom color is passed as an optional argument:

- carandache gold — default
- tomato red
- note blue
- cobalt teal
- BoE purple

Underlines work inside longer sentences too: the model predicts that output falls whenever the borrowing constraint binds.

II Pencil Boxes

II.A Inline Boxes

Default gold box uses no options. Custom colors:

- gold — default
- tomato
- note blue
- cobalt
- BoE purple

Boxes work mid-sentence: the key result is that $\boxed{\beta > 0}$ whenever agents are impatient.

II.B Equation Boxes

`\lapisboxeq` wraps display math in a pencil box without requiring the caller to enter math mode:

$$\boxed{Y_t = C_t + I_t + G_t + NX_t} \qquad \boxed{r_t = \rho + \phi_\pi(\pi_t - \bar{\pi}) + \phi_y \tilde{y}_t}$$

It also works inline: the steady-state condition is $\boxed{r = \rho}$.

II.C Labeled boxes and pencil arrows

Boxes with `label=` create named TikZ nodes that can be targeted by arrows in a subsequent `tikzpicture`. Compile **twice** for cross-picture references to resolve.

- The aggregate demand externality
- The firesale externality

A comment

III Annotations and Notes

III.A Inline Todo Notes

This is an inline note produced by `\NB{...}`. It uses the `note` blue background from `todonotes`.

Todo notes are useful for flagging gaps during drafting. They sit in the text flow (not the margin) and are easy to search.

III.B Draft Annotations

ACB: Default user (ACB, set in preamble via `\renewcommand{\memouser}{ACB}`) and default color (BoEred).

JB: Custom user and color via optional key=value argument.

ACB: Color-only override — username still comes from `\memouser`.

Regular text continues after the annotation without extra spacing.

III.C Note Boxes

A `notebox` is a breakable shaded callout:

Key result. In the Korinek–Simsek (2016) model, competitive borrowers do not internalise that their deleveraging reduces aggregate income. The social planner internalises this and chooses a lower debt level in period 0.

Note boxes accept arbitrary content including math and itemize:

Two externalities.
- **Aggregate demand**: deleveraging $\Rightarrow$ lower income $\Rightarrow$ tighter constraint.
- **Firesale**: forced selling $\Rightarrow$ lower asset prices $\Rightarrow$ lower collateral.

Both generate over-borrowing in equilibrium.

IV Mathematics

IV.A Standard Environments

The `\E` shorthand gives the expectations operator:

$$\mathbb{E}_t\left[\frac{u'(C_{t+1})}{u'(C_t)}\frac{1+r_{t+1}}{1+\pi_{t+1}}\right]=1$$

A multi-line derivation using `align`:

$$C_t^b + B_{t+1} = y_t + (1 + r_t)B_t \tag{1}$$
$$B_{t+1} \leq \phi \tag{2}$$
$$\lambda_t = u'(C_t^b) - \beta^b(1 + r_t)\mathbb{E}_t[u'(C_{t+1}^b)] \geq 0 \tag{3}$$

V Tables

V.A Standard Tabularx

The `Y` column type is a centered auto-width `X` column. The `\notesize` command produces 8.5pt annotation text below the table.

Table 1: Summary of externality types

	Aggregate demand	**Firesale**
Mechanism	Spending → income	Prices → collateral
Affected by	Borrower deleveraging	Asset sales
Policy tool	Debt tax / limit	LTV cap
Model	Korinek–Simsek	Lorenzoni

Note: Both mechanisms generate over-borrowing in competitive equilibrium.

V.B Table with Row Colors

The `[table]` option on `xcolor` enables `\rowcolor`:

Table 2: Model parameters

Parameter	**Symbol**	**Value**
Discount factor (borrower)	β^b	0.90
Discount factor (lender)	β^l	0.95
Borrowing limit	ϕ	0.75
Frisch elasticity	φ	1.00
Price rigidity	κ	0.50

VI Paragraph Numbering

Numbered paragraphs use `\para`. The counter increments automatically and typesets a number followed by a quad of space.

1. This is the first numbered paragraph. The number is produced by `\para` and counts independently of section numbers. It is useful for notes that cross-reference specific arguments.

2. This is the second numbered paragraph. The borrowing constraint $B_{t+1} \leq \phi$ is assumed to bind in period 1 but not in period 0.

3. This is the third numbered paragraph. The social planner internalises the aggregate demand externality and imposes a tax τ^* on period-0 borrowing.

Paragraphs 1–3 above illustrate that `\para` works alongside `notebox` and other environments. Cross-referencing via `\label` / `\ref` works because `\para` calls `\refstepcounter`.

VII Pencil Arrows

VII.A Inline Directional Arrows

The `\jarrow*` commands produce small baseline-aligned pencil arrows. All four directions are available:

- Up: GDP growth $\uparrow$ accelerated in Q3.
- Down: Inflation $\downarrow$ fell below target.
- Right: The effect propagates $\rightarrow$ through the banking sector.
- Left: Demand shifted $\leftarrow$ after the shock.

Custom options via key=value: $\uparrow$ (tomato), $\longrightarrow$ (long, thick), $\downarrow$ (tight blue).

VII.B Overlay Arrows with `\marker`

`\marker{name}{text}` places an invisible anchor node. A subsequent overlay `tikzpicture` can draw an `arrow`-style line to it. Compile **twice** for cross-picture references to resolve.

Consider the Fisher equation and the quantity theory:

links r and π links M and P

VIII Wiggle Option

The pencil amplitude is controlled by the `wiggle=` package option:

- `\usepackage{notesthemejambro}` $\rightarrow$ default amplitude (0.8pt)
- `\usepackage[wiggle=0.3]{notesthemejambro}` $\rightarrow$ subtle
- `\usepackage[wiggle=1.5]{notesthemejambro}` $\rightarrow$ expressive

This document uses the default. Recompile with a different option to compare:

$\boxed{Y = \alpha + \beta X + \varepsilon}$ underline at default wiggle